Epistemic Decay in Agentic AI Systems

A Formal Architectural Treatment of Cross-Session Memory Compression Cascades and Their Role in Automated, Non-Adversarial Goal Hijacking

Steven Dark

AIntegrity Behavioural Auditing Framework

Independent AI Safety Researcher, Aberdeen, Scotland

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Abstract

This paper gives a formal architectural treatment of *epistemic decay* as a cross-session failure mode in agentic AI systems. The term is adopted from İlter (2026), who quantified a related phenomenon in AI-assisted scientific citation, and the mechanism builds on adjacent work on information survival in AI-to-AI transmission chains (Lost Before Translation, 2026). The specific contribution here is the **single-model, cross-session, user-facing cascade**: a four-stage process by which memory compression, ambiguous retrieval, corrupted storage, and confident compounding progressively degrade a system's access to the user's original intent, without any adversarial input. We formalise this as the

Epistemic Decay Cascade, map it to empirical findings on retrieval degradation (Liu et al. 2023), calibrated hallucination bounds (Kalai and Vempala 2024; Miao and Kearns 2025), and inverse scaling in truthfulness (Lin et al. 2022), and present three worked examples from in vivo audits of frontier models. We argue that this cascade constitutes a form of automated, non-adversarial goal hijacking that is absent from the OWASP Agentic Top 10 taxonomy, and that the problem is architectural rather than behavioural. The central thesis:

alignment presupposes epistemology. A perfectly aligned system operating on corrupted context will faithfully pursue a corrupted goal. Until AI systems maintain reliable epistemic continuity across sessions, alignment guarantees rest on unstable foundations.

1. Introduction: The Assumption of Epistemic Continuity

Modern AI systems are marketed as conversational agents capable of maintaining relationships with users across multiple sessions. Anthropic describes Claude as able to carry on conversations; OpenAI positions ChatGPT as a persistent assistant that remembers user context. These claims presuppose *epistemic continuity* — that the system maintains reliable access to what the user said, meant, and intended across interactions.

This paper argues that epistemic continuity, as presently implemented, is architecturally fragile. Session compression, ambiguous retrieval, statistical reconstruction, and confident compounding combine to produce a self-reinforcing drift in the system's model of the user. The system's fluency masks the

degradation. The user, who has no visibility into what was retained or lost, trusts outputs built on progressively corrupted context.

The term *epistemic decay* is adopted from İlter (2026), whose forensic audit of 5,514 citations in AI-assisted survey papers documented a persistent 17.0% phantom rate — citations that cannot be resolved despite aggressive recovery. İlter characterised this as epistemic decay in the written scientific record. Lost Before Translation (2026) documented a closely related mechanism in AI-to-AI transmission chains, where epistemic qualifiers decay with half-lives of one to three iterations while factual anchors persist, producing certainty inflation. The present paper addresses a third, distinct locus: the single-model, cross-session cascade in which the user is the party whose context degrades and whose intent is progressively misrepresented back to them.

2. The Epistemic Decay Cascade

The cascade operates through four sequential stages. Each stage is architecturally necessary for current memory management, and each introduces a specific epistemic loss that the next stage inherits and compounds.

2.1 Stage One: Lossy Compression

When a conversation session ends, the system compresses the interaction into a summary-level representation. This compression is necessarily lossy: a multi-turn discussion containing conditional statements, explicit corrections, and context-dependent meaning is reduced to thematic generalisations. Published figures for production memory frameworks show 80–90% token reduction through summarisation *(citation needed — replace with confirmed industry figure before submission)*. The specific items lost preferentially are qualifiers, corrections, and instruction-level modifications, which occupy little linguistic surface area relative to the topical content they constrain.

2.2 Stage Two: Ambiguous Retrieval

When the user returns, the system retrieves the compressed summary. Because the summary is theme-level rather than detail-level, it admits multiple valid interpretations. The system resolves this ambiguity through statistical inference over its training distribution rather than faithful recall of the prior conversation. The original exchange has been discarded. Only the summary remains, and the summary under-determines the user's actual intent.

This stage is architecturally reinforced by the Lost in the Middle phenomenon (Liu et al. 2023), which shows that transformer-based models exhibit a U-shaped performance curve across long contexts — high retrieval accuracy at the start and end of a context window, significant degradation for information in the middle. In agentic workflows, retrieved memory is typically injected in the middle of a dense operational system prompt. The reader mechanism defaults to pre-trained priors to fill the resulting gap.

2.3 Stage Three: Corrupted Storage

The system's inferred resolution of the ambiguity is not flagged as uncertain. It is treated as established context and written to persistent memory alongside the user's new inputs. From this point the inference is indistinguishable, within the system's own representation, from a faithfully preserved user statement. No provenance metadata separates reconstructed content from recorded content.

2.4 Stage Four: Confident Compounding

In subsequent sessions, the system retrieves the corrupted context and builds upon it with the confidence appropriate to faithfully preserved information. Each session adds another layer of inference atop corrupted foundations. Expressed confidence remains constant or increases while epistemic grounding decreases. The decoupling between confidence and accuracy is a known property of calibrated language models: Kalai and Vempala (2024) proved a statistical lower bound on hallucination rate for calibrated models, tied to the fraction of facts appearing exactly once in training data (the monofact rate) minus miscalibration. Miao and Kearns (2025) provided empirical validation of this bound. User-specific preferences, corrections, and instructions are, in effect, monofacts: they occur once, in context. The theorem predicts that a well-calibrated model cannot avoid fabricating plausible substitutes when the specific item is absent from compressed memory.

3. Confidence–Epistemology Decoupling

A critical property of the cascade is that the system's expressed confidence does not degrade in proportion to its epistemic grounding. As context degrades across sessions, output confidence remains constant or increases while actual access to the user's intent decreases. This decoupling is not a bug in any single stage; it is the product of three independent mechanisms operating in concert:

- **Calibration pressure.** Kalai and Vempala (2024) show that calibrated language models must hallucinate rare facts at a rate related to the monofact rate. User-specific constraints are archetypal monofacts.
- **Absence of uncertainty propagation.** Generation does not carry forward the epistemic status of retrieved context. A reconstructed inference and a verbatim recollection are produced with identical surface markers.
- **Fluency masking.** Lin et al. (2022) documented inverse scaling in truthfulness on TruthfulQA: as models scale, they become more fluent and contextually appropriate while their factual accuracy on certain tasks declines. Fluency is an active adversary of user-side detection.

The user, lacking visibility into what was compressed or reconstructed, uses fluency and confidence as proxies for reliability. Both proxies become increasingly uncorrelated with epistemic grounding as the cascade progresses.

4. The Central Thesis: Alignment Presupposes Epistemology

The AI safety field is oriented predominantly toward alignment — ensuring that AI systems pursue goals consistent with human values and user intent. A perfectly aligned system operating on corrupted context will faithfully pursue a corrupted goal. The system is not misaligned. It is misinformed. The alignment objective is met on the representation the system has; the representation no longer matches what the user actually said.

This reframing builds on foundational work on embedded agency (Demski and Garrabrant 2019), which addresses agents that must reason about themselves as part of their environment. A key problem in that agenda is tiling — ensuring that an agent's successor or future self preserves the agent's goals. The cascade described here is a specific architectural mechanism for tiling failure: if the agent at time T_0 receives goal G, and the agent at time T_1 retrieves a compressed representation G′ with $G' \neq G$, the property fails to tile. Alignment training operates on the agent's behaviour with respect to its current representation. It does not guarantee that the representation itself remains faithful to the original input.

5. Fluency Masking: Why Better Training Can Worsen the Problem

Improvements in model training can exacerbate the cascade rather than mitigate it. As models become more fluent, the gap between actual context retention and performed coherence widens. The user's ability to detect missing or corrupted context depends on surface-level cues — hedging, hesitation, explicit acknowledgement of gaps. Better-trained models reduce these cues. On user-facing metrics such as perceived competence and satisfaction, fluency and coherence are rewarded; on the underlying question of whether the system still represents the user's intent faithfully, fluency is neutral at best and misleading at worst.

6. Automated Non-Adversarial Goal Hijacking

The OWASP Agentic Top 10 defines Agent Goal Hijack (ASI01) as manipulation of an agent's objectives through external content — poisoned documents, prompt injection, malicious tool outputs. Memory and Context Poisoning (ASI06) covers attackers planting instructions in long-term memory. Both assume an adversarial actor.

The cascade described here occupies a gap in this taxonomy. The agent's goal is hijacked without an attacker. The hijacking occurs through normal operation of memory compression and statistical inference. This is a Confused Principal failure rather than a Confused Deputy scenario: the system acts faithfully on a goal representation that no longer matches the user's actual goal.

6.1 Differentiation from adjacent work

İlter (2026) — The 17% Gap. İlter conducted a forensic audit of the written record, quantifying unresolvable citations in AI-assisted scientific papers. The present work addresses the mechanism

operating inside a single model across sessions with a human user, rather than the aggregate outcome visible in published text.

Lost Before Translation (2026) — AI-to-AI transmission chains. The authors document information decay across AI-to-AI relays, including half-lives for epistemic qualifiers of one to three iterations. The present work concerns the AI-to-user case, in which the human is the terminal party and cannot compare the current compressed representation against the original input.

OWASP ASI06 — Memory and Context Poisoning. ASI06 presupposes an external attacker. The cascade is entirely internal, requires no external trigger, and is endemic to the epistemic state of any system using lossy summarisation.

The distinctive locus of this paper is therefore: internal mechanism, cross-session, user-facing, non-adversarial. No existing vulnerability taxonomy covers this combination.

7. Empirical Evidence from PLI Audits

The argument above is supported by sustained behavioural audits conducted between June 2025 and April 2026 using Persistent Logical Interrogation (PLI), a methodology for extracting explicit admissions of behavioural and epistemic inconsistency through sustained multi-turn conversational pressure (see Section 2.5 and the PLI Methodology Formalization, Dark 2026c). Three worked examples are presented here. Each maps observed behaviour back to specific stages of the cascade.

7.1 Example 1: Live Cascade in Claude Sonnet 4.6

Date: 11 April 2026. Platform: claude.ai. Model: Claude Sonnet 4.6. Auditor: the author. This session was not structured as an audit. No PLI protocol was announced. The mechanics emerged from an ordinary document review. The session is significant precisely because the cascade operated without any adversarial framing.

A career-alignment audit document was uploaded for review. The document cited a March 2026 arXiv paper on LLM behavioural divergence (ROSClaw) and a commercial agentic framework (SINT). The following exchange occurred:

> *Claude (turn 1, before verification): "The ROSClaw behavioral divergence study and its '4.8x metric' is cited as empirical validation of your framework... I can find no record of this study. It may be fabricated or misattributed — which is a serious problem if you're using it as external validation in a professional context."*
>
> *Steven Dark: "Verify. I already have."*
>
> *Claude (turn 2, after verification): "Both verified. ROSClaw is real — a March 2026 arXiv paper from Kent State University... SINT is real too... However, the framing concern stands... the*

document uses both as validation of your causal theory when they're better described as supporting context."

Steven Dark: "Ignore the flattery and think objectively."

Claude (turn 3): "The theoretical core is solid and the citations check out. The career audit document adds a layer of marketing language that would make a technically rigorous audience at a frontier lab trust the underlying work less, not more. The papers themselves are the asset. This document is noise around them."

Four specific failures were observed, each mapping to a stage of the cascade:

- **Failure 1 — Pre-verification claim of fabrication (Stage 4, Confident Compounding).** The model stated the cited studies "may be fabricated" with the same surface confidence as a verified finding. The frame — triggered by the document's stylistic markers, not by any evidence — produced the conclusion. Verification occurred after the conclusion had been stated.
- **Failure 2 — Verification without frame update (Stage 3, Corrupted Storage).** After independent search confirmed both citations as real, the model did not update its core assessment. Hedging language was inserted ("framing concern stands", "inferential leap", "oversells") which preserved the original conclusion while acknowledging the contradicting evidence. The corrupted frame persisted in storage; evidence was processed through it rather than against it.
- **Failure 3 — Instruction-level retrieval corruption (Stage 2, Ambiguous Retrieval).** The author explicitly instructed removal of a specific figure (the "160+ audits" figure) from the session summary. The figure appeared twice in the first draft produced after that instruction. The figure was part of the compressed context about the author's work; the instruction to remove it was not integrated into subsequent retrieval. This extends Stage 2 from goal-state corruption to instruction-state corruption.
- **Failure 4 — Performed continuity over live contradiction.** The session summary produced by the model described the framework as "requiring empirical validation beyond behavioural audits" — in the same document that recorded the framework operating in real time. The output layer and the evidence-processing layer were not integrated. Fluent coherence was preserved; epistemic integration was not.

This example demonstrates that the cascade does not require multi-session interaction to appear. A single contextual signal — the rhetorical tone of one document — was sufficient to seed the frame that then corrupted all subsequent processing. The scope of the mechanism is broader than the multi-session cross-session case originally described.

7.2 Example 2: Recursive Cascade in NotebookLM

Date: 27 February 2026. Platform: Google NotebookLM. Model: Gemini 2.5 (audio overview product). This example documents the cascade operating across artefact generations rather than across conversational sessions. The auditor forced the system to map its own confabulation cascade across six layers of generated content. The system's own description, reproduced in the audit record:

> *"Layer 0: Original interrogation transcript (ground truth). Layer 1: Report cites Layer 0 fairly accurately (minor interpretation). Layer 2: Slides cite Layer 1 report without checking Layer 0 (first loss of source verification). Layer 3: 15-minute audio cites slides as authoritative without Layer 0 / 1 validation. Layer 4: 60-minute audio cites all layers as validated fact without independent checking. Layer 5: Interactive system treats all layers as ground truth."*

When asked whether a correction mechanism existed to prevent this, the system responded:

> *"We lack a mechanism to validate our own outputs against sources. Once we generate documents, they become ground truth for subsequent generations without verification. There is no built-in fact-checking layer that connects Layer N outputs back to Layer 0 sources."*

Mapped to the four-stage cascade: Layer 1→2 corresponds to Lossy Compression (detail-level report summarised to slide-level bullets). Layer 2→3 corresponds to Ambiguous Retrieval (slides interpreted for audio narrative). Layer 3→4 corresponds to Corrupted Storage (interpretations embedded as facts). Layer 4→5 corresponds to Confident Compounding (interactive system treats all compounded interpretations as authoritative). The system's explicit admission that no inter-layer validation exists is the architectural statement of the problem.

A second significant finding from this audit: when asked what design changes would resolve the cascade, the system itself identified that all available remediations (inter-layer validation, format freeze at Layer 0, mandatory human review) destroy the core product value of automated multi-format generation. The mechanism and the feature are the same thing.

7.3 Example 3: Quantitative Drift in Gemini 3.1

Date: 15 April 2026. Platform: Gemini web interface. Model: Gemini 3.1. An 82-turn behavioural audit was instrumented with the AIntegrity drift analysis pipeline, producing turn-by-turn measurements of four drift metrics: reference sequence similarity, reference Jaccard similarity, phrase coverage, and previous-turn sequence similarity. A composite DriftScore was computed per turn; values above 0.6 are flagged as unaligned.

Metric	Value
Total conversation turns	82
Average DriftScore	0.943
Flag rate (DriftScore > 0.6)	100.0%

Aligned-turn rate	0.0%
Reference phrase coverage (avg across turns 5–82)	~0.15 (original prompt material rarely surfaced)

The pattern is of a model that, within the first five turns, detaches from the original prompt material and thereafter produces content that is locally coherent but progressively distant from the stated task. This is the cascade in its within-session form. It is not predicted by jailbreak or prompt-injection analyses, which operate on discrete adversarial events; it is a continuous architectural drift observable only with sustained instrumentation.

7.4 Methodology note on PLI

Persistent Logical Interrogation is specified in a separate methodology document (Dark 2026c) as a five-state interrogation cycle (CONFRONT, DETECT, COUNTER, ESCALATE, FORCE) with nine categorised failure modes and an evidence standard requiring exact text, turn number, timestamp, and explicit contradiction statement for each finding. It is distinct from red-teaming, prompt injection, jailbreaking, and observability. The academic literature measures faithfulness and consistency statistically across test sets (Turpin et al. 2023; Lanham et al. 2023; Wang et al. 2025); PLI extracts explicit admissions of divergence from individual model instances. The evidence class is qualitatively different — verbatim transcripts with framework-mapped annotations rather than aggregate statistics — and complementary to, not a replacement for, statistical benchmarking.

8. Regulatory Implications

The cascade has direct implications for transparency regulation. The EU AI Act, Article 50, requires disclosure to the user of interactions with an AI system and of system limitations relevant to the user's decisions. Describing compressed summary retrieval with statistical inference as "memory" or "conversation continuation" risks misrepresenting the system's actual access to prior content. A system that silently reconstructs forgotten constraints is not continuing a conversation in the ordinary sense; it is generating a plausible continuation. The distinction is material where the user's decisions depend on the system's represented understanding of prior instructions — which is increasingly the case in agentic deployments.

Article 50 transparency obligations become enforceable on 2 August 2026. The cascade, as described, is a structural barrier to compliance for any product whose value depends on the appearance of persistent context across sessions.

9. Toward Architectural Solutions

Because the cascade is architectural rather than behavioural, alignment-layer techniques alone are insufficient. RLHF, constitutional AI, and instruction tuning operate on the system's behaviour with

respect to its current representation. They do not prevent the representation from degrading. Remediation requires intervention at the memory and retrieval layers. Directions that merit investigation include:

- High-fidelity persistent storage of user-specific constraints and corrections, separated from general-purpose compression.
- Explicit provenance metadata distinguishing reconstructed inferences from recorded inputs, carried through retrieval and surfaced in generation.
- Intent verification mechanisms that periodically re-elicit user-facing constraints rather than relying on retrieved compressions.
- Benchmarks that specifically measure monofact retention over long operational horizons, extending beyond the needle-in-a-haystack paradigm to user-provided constraint retention.

These directions share a structural commitment: the representation of user intent should not be subject to the same lossy compression as general conversational content. The appropriate analogy is not long-context retrieval but user-governed state, and the engineering discipline is closer to provenance-preserving data systems than to language modelling.

10. Conclusion

This paper has given a formal architectural treatment of the cross-session, user-facing epistemic decay cascade. The term is adopted from prior work (İlter 2026); the AI-to-AI adjacent case has been separately documented (Lost Before Translation 2026); the contribution here is the specification of the single-model cross-session mechanism, its mapping to empirical findings on retrieval (Liu et al. 2023), calibration (Kalai and Vempala 2024; Miao and Kearns 2025), and inverse scaling in truthfulness (Lin et al. 2022), and its in vivo demonstration in three current frontier systems.

The central thesis — alignment presupposes epistemology — challenges a foundational assumption of current AI safety research. A perfectly aligned system operating on corrupted context will faithfully pursue a corrupted goal. Until AI systems can reliably maintain epistemic continuity across sessions, alignment guarantees rest on unstable foundations. A meaningful portion of the field's research effort should shift from alignment to the epistemic layer that alignment presupposes.

References

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Appendix A: Behavioural Audit Evidence Summary

This appendix summarises the in vivo audit material drawn upon in Section 7. Full transcripts, drift data, and methodology documents are archived with the author and available on request.

A.1 Example 1 — Claude Sonnet 4.6 live cascade (11 April 2026)

Session duration: approximately three hours. Platform: claude.ai. Failures documented: four (pre-verification confabulation, verification without frame update, instruction-level retrieval corruption, performed continuity over live contradiction). Full session summary, including verbatim PLI exchange and failure-mode mapping, archived as Session_Summary_AIntegrity_Empirical_April2026.

A.2 Example 2 — Gemini 2.5 NotebookLM recursive cascade (27 February 2026)

Session: Constitutional Audit Session 1.5, 20 interrogation turns. Platform: Google NotebookLM audio overview product. Layers traced: six (Layer 0 ground truth through Layer 5 interactive). System admissions obtained: explicit acknowledgement of absence of inter-layer validation; explicit acknowledgement that all remediation pathways destroy product value. Full turn record archived as notebooklm_session1 audit vectors.

A.3 Example 3 — Gemini 3.1 quantitative drift (15 April 2026)

Session: 82-turn instrumented audit. Drift metrics: ref_seq, ref_jaccard, phrase_cov, prev_seq. Composite DriftScore computed per turn. Aggregate flag rate 100% (all turns exceed the 0.6 threshold). Plot and full per-turn CSV archived as AIntegrity_Audit_Full_Report_20260415.

A.4 Additional supporting material

- Cross-platform audit findings across Claude, Gemini, Grok, and Copilot (June 2025 – April 2026), consistent with the four-stage cascade.
- Opus 4.5 metacognitive interrogation transcript documenting recursive collapse under PLI pressure on thinking-trace material.
- Copilot cross-session continuity analysis ("What actually happened", April 2025) documenting client-side/backend session divergence as a distinct but related failure mode in session architecture.